

# Syllabus



This course investigates the fundamental issues addressed by scientists studying tropical rain forests, including what gave rise to their remarkable biodiversity, the drivers and consequences of deforestation, why people are fascinated by rain forests, cultural and ecological stereotypes about the tropics, and different approaches to forest conservation and sustainability. ***By the end of the course students will be able to:***

- Recognize and describe stereotypes about rain forests & their residents
- Analyze rain forest tropes in art, literature, & popular culture
- Discuss & evaluate hypotheses for the origins and maintenance of tropical biodiversity
- Explain & compare the history and ecological influence of humans in rain forests
- Review contemporary threats to rain forests
- Analyze and visualize data on deforestation
- Review and contrast strategies for rain forest conservation & restoration
- Identify rain forests in their daily lives & set personal goals for advancing their conservation
- Produce materials for communicating about tropical biology & conservation to family and peers

## When & Where

**When:** Tuesdays 11:45 AM-12:35 PM (Period 5) & Thursdays 11:45 AM-1:40 PM (Periods 5-6)

**Where:** Larsen Hall 0310 (both days)

## Instructor & TA

**Instructor:** Dr. Emilio M. Bruna (email: [embruna@ufl.edu](mailto:embruna@ufl.edu), phone: (352) 846-0634)

## Credits & Prerequisites

**Credits:** 3

**Prerequisites:** None

**Quest Program:** Quest 2

**GenEd Designation:** International

A minimum grade of C is required for Quest and General Education credit. Courses intended to satisfy Quest and General Education requirements cannot be taken S-U.

## Minors & Certificates

This course counts towards the Minor in Latin American Studies. See <https://www.latam.ufl.edu/academics/undergraduate-programs> for more information.

# Required Materials

**Materials and Supplies Fees:** None.

**Textbooks to purchase:** None.

## ! Important

All materials, including readings and videos, will be made available on the course Canvas page. *Students should sign up for free online access to the New York Times, Wall Street Journal, and The Economist by following the instructions at [this UF Libraries Website](#)<sup>1</sup>.*

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<sup>1</sup>Some of the assigned readings from these sources have dynamic multimedia data visualizations or video that can't be appreciated in the .pdf versions posted to canvas.

# Office Hours

💡 Can you give me **one good reason** why I should go to Office Hours?

I can give you **ten**.

1. To introduce yourself.
2. Get clarification on assignments.
3. Argue about a topic that came up in class.
4. Grab a (free) tea, coffee, or espresso in our lab kitchen.
5. Make sure you understood the key points from a class session.
6. Ask for feedback on ideas for course projects.
7. Get advice on successfully navigating academic life at UF.
8. Discuss how to gain experience for your post-graduation goals.
9. Request help arranging a study group.
10. You don't need a good reason...just come on by.

## When

- **Instructor:** Wednesday 1:30-3:30 (in-person and via Zoom, see below for details). You can drop by any time during the session, but if you want to guarantee a specific time slot you can do so by signing up here: <https://embruna.youcanbook.me/>. If you can't make it during the regularly scheduled office hours, you can send a message via Canvas to set up an appointment.
- **Teaching Assistant:** Tuesday 4:00-5:30 pm (in-person & online) or by appointment.

## Where

- **In-person:** The Tropical Ecology & Conservation Lab. The lab is located at 711 Newell Drive next to Rawlings Hall (between the bus stop and the parking garage). For directions you can use this [link to the lab on Google Maps](#).
  - **Online:** To join Office Hours virtually use the Zoom Conferences link on the main menu of the course Canvas page. We are online the entire session.
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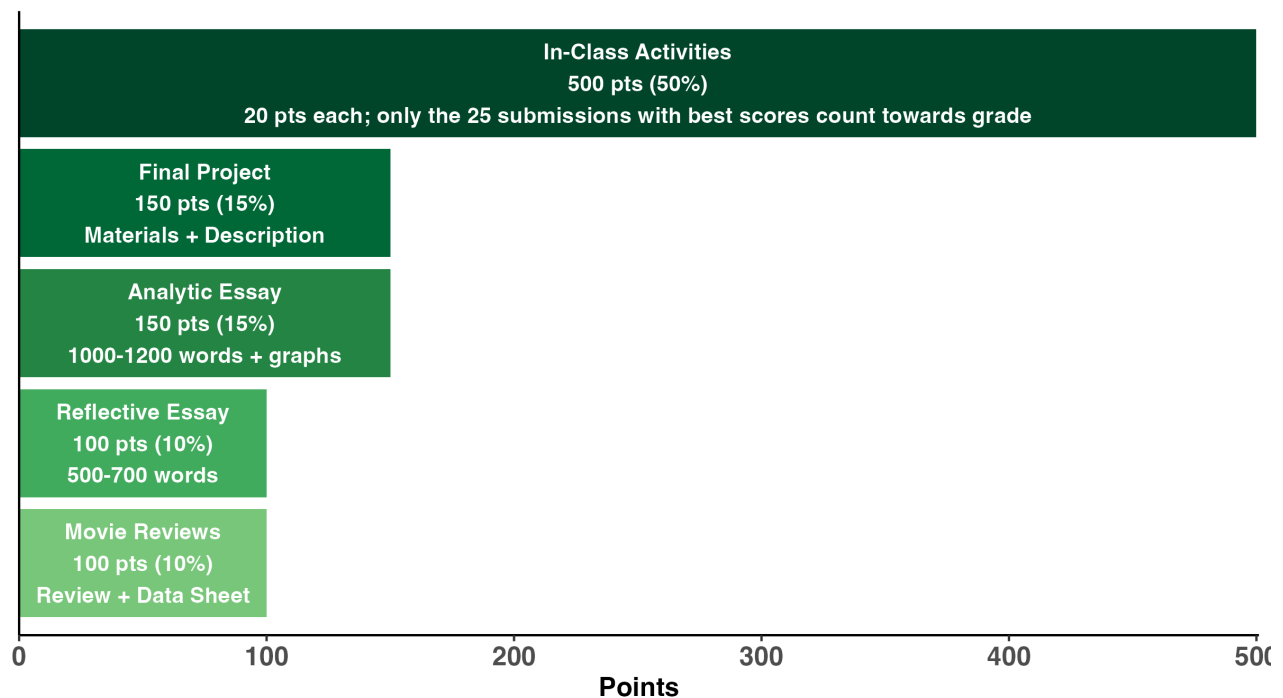
# Grades & Attendance

Learning in our class is achieved with an diverse array of methods ranging from data analysis to essays to projects. In most class sessions you will also be working with small groups of students to complete an in-class assignment that reinforces the major themes of the day's topic. In keeping with the philosophy of the Quest program, this course also has *Experiential Learning and Self-Reflection Components*.

💡 Important note regarding class discussions & group work.

We will explore some challenging, important problems and increase our understandings of different perspectives and approaches for addressing them. These conversations may not always be easy; we sometimes will make mistakes in both how we communicate our perspective and what we hear other say. There may be times when we need patience, courage, imagination, and of course mutual respect to engage our texts, classmates, instructors, guests, and our own ideas and experiences. *Disrespectful or disruptive behavior will not be tolerated*. And always remember that as scholars we rely on critical thinking, data, prior scholarship, and expert opinion when interrogating assigned readings and discussing course content with classmates and instructors.

## Assignments (1000 pts)



## Grading

**In-class Assignments are due by the following class session.** Late assignments will lose 10 pts.

**Regrades:** Requests for re-evaluation of assignments must be accompanied by an explanation for why you think you deserve additional credit and the number of additional points you think you deserve. The deadline for submission is one week after the work was returned.

## Grades & Attendance

**Grade Assignment** (based on % of total points earned): A = 94–100%, A- = 90–93%, B+ = 87–89%, B = 84–86%, B- = 80–83%, C+ = 77–79%, C = 74–76%, C- = 70–73%, D+ = 67–69%, D = 64–66%, D- = 60–63%, E < 60

**Grade Points:** For information on how UF assigns grade points, visit: <https://catalog.ufl.edu/UGRD/academic-regulations/grades-grading-policies/>

### Attendance

Though attendance is not required, many of the sessions we will be completing activities in class that count towards your grade. Some these can be completed independently, but by doing them in class you will benefit from working collaboratively with the other students. **Not all in-class activities can be completed independently.** That is why only a subset of the in-class assignments count towards your grade. We also offer some opportunities for extra credit. ***If you need to miss class for any reason, please let me know as soon as possible.*** We will try to make arrangements for you to complete any assignments and go over any material you will be missing.

### Participation

Consistent informed, thoughtful, and considerate class participation is encouraged (and in some cases required). *If you have personal issues that prohibit you from joining freely in class discussion (e.g., shyness, language barriers, medical condition): no problem.* let us know and we will discuss alternative modes of participation.

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## Topic Calendar & Key Dates

Table 0: Course Topics and Key Dates

| V                                                           | DATE   | TOPIC                                                | ASSIGN-<br>MENT INFO         |
|-------------------------------------------------------------|--------|------------------------------------------------------|------------------------------|
| Wk<br>1                                                     | 20-Aug | Course Welcome and Introduction                      |                              |
| <b>WHY ARE WE FASCINATED BY TROPICAL RAIN FORESTS?</b>      |        |                                                      |                              |
| Wk<br>2                                                     | 25-Aug | Historical Narratives                                |                              |
| Wk<br>3                                                     | 27-Aug | Historical Narratives, cont.                         |                              |
|                                                             | 1-Sep  | Rain forests in Art & Literature                     |                              |
|                                                             | 3-Sep  | Rain forests in Pop Culture                          | Movie Reviews<br>Assigned    |
| <b>THE ECOLOGY &amp; EVOLUTION OF TROPICAL RAIN FORESTS</b> |        |                                                      |                              |
| Wk<br>4                                                     | 8-Sep  | What is a rain forest?                               |                              |
|                                                             | 10-Sep | Where is a rain forest?                              |                              |
| Wk<br>5                                                     | 15-Sep | Patterns of biodiversity                             |                              |
|                                                             | 17-Sep | Patterns of biodiversity, cont.                      |                              |
| Wk<br>6                                                     | 22-Sep | Origins of tropical biodiversity                     |                              |
|                                                             | 24-Sep | Maintenance of tropical biodiversity                 |                              |
| Wk<br>7                                                     | 29-Sep | Humans in rain forests                               |                              |
|                                                             | 1-Oct  | Humans in rain forests, cont.                        | Movie Reviews<br>Due         |
| Wk<br>8                                                     | 6-Oct  | Paradox of Luxuriance & (Bio)Narratives<br>Revisited |                              |
|                                                             | 8-Oct  | JUNGLE FILM FESTIVAL                                 |                              |
| <b>THE DRIVERS AND IMPACTS OF DEFORESTATION</b>             |        |                                                      |                              |
| Wk<br>9                                                     | 13-Oct | How much rain forest is there?                       | Analytic Essay<br>Assigned   |
|                                                             | 15-Oct | How much rain forest have we lost?                   |                              |
| Wk<br>10                                                    | 20-Oct | Drivers of deforestation                             |                              |
|                                                             | 22-Oct | Drivers of deforestation, cont.                      |                              |
| Wk<br>11                                                    | 27-Oct | Rain forests & climate                               | Analytic Essay<br>Due        |
|                                                             | 29-Oct | Rain forests & climate change                        | Reflective Essay<br>Assigned |
| <b>THE FUTURE OF TROPICAL RAIN FORESTS</b>                  |        |                                                      |                              |
| Wk<br>12                                                    | 3-Nov  | International conservation frameworks                |                              |
|                                                             | 5-Nov  | Trade in tropical commodities & DURIAN<br>FEST       |                              |
| Wk                                                          | 10-Nov | Consumer choices & rain forest conservation          |                              |

# FAQ

**How do I contact the Instructor or TA?** Send a message in CANVAS. That way both the TA and instructor will see it and you will get a response more quickly (emails sent to us directly tend to get buried under other messages).

**How will you send class announcements?** CANVAS. Check the course page for announcements and be sure you are receiving Canvas emails and updates.

**What work should we complete BEFORE class?** Read, watch, listen to, or review all materials assigned for the session. This material sets the stage for in-class activities.

**What will we do DURING class?** In-class exercises that reinforce key concepts, discussions of the assigned readings, and let you practice skills in other assignments. Some are completed individually, while others require working in groups or pairs. Each activity will have instructions and a rubric; most are designed to be finished in class.

**When is the “in-class” work due?** Deadlines are in Canvas, but typically the start of the next class session.

**What if I miss class?** Attendance is not required, but in many of the sessions we will be completing activities in class that count towards your grade. Some of these can be completed independently, but by doing them in class you will benefit from working collaboratively with the other students. However, some of the in-class activities can not be completed on your own. That’s why only a subset of the in-class assignments count towards your grade and we offer extra credit.

**I have to miss class on a certain date. What should I do?** Let us know as soon as possible so we can make arrangements for you to review material you will miss and complete assignments.”)

**Class discussions are difficult for me. Will this affect my grade?** No! If there are issues that make engaging in discussions difficult (e.g., shyness, language barriers, a medical condition), let us know and we will find alternative modes of participation.

**I unexpectedly have no child care today...what can I do?** UF does not have a policy on children in the classroom, so here is mine: I never want students to feel they have to choose between their responsibilities as parents and their education. Bringing your child to class because of a gap in child care is totally fine (and that includes babies). If you do so, please try and sit close to the exit so that you can more easily step outside if you need to care for them and so other students can continue learning.



# UF Policies

## Student Accommodations

Students with disabilities or learning barriers that would like to request academic accommodations should connect with the Disability Resource Center by visiting <https://disability.ufl.edu/students/get-started/>. Please share your letter with me and discuss access needs as early as possible in the semester so that I can do whatever is necessary to ensure your participation and learning.

## Course Evaluations

Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations. Guidance on how to provide constructive feedback is available at <https://gatorevals.aa.ufl.edu/students/>. Students will be notified when the evaluation period opens. Summaries of course evaluation results are available to students at <https://gatorevals.aa.ufl.edu/public-results/>. Students can complete evaluations in three ways: (1) The email they receive from GatorEvals, (2) Their Canvas course menu under GatorEvals, and (3) The central portal at <https://my-ufl.bluera.com>.

## University Honesty Policy

UF students are bound by The Honor Pledge which states, “We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: On my honor, I have neither given nor received unauthorized aid in doing this assignment. The Honor Code (<https://www.dso.ufl.edu/sccr/process/student-conduct-honor-code>) specifies a number of behaviors that are in violation of this code and the possible sanctions. Furthermore, you are obligated to report any condition that facilitates academic misconduct to appropriate personnel. If you have questions or concerns, consult with the instructor or TAs.

## Software Use

All faculty, staff & students are required & expected to obey laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against UF policies & rules, disciplinary action will be taken as appropriate.

## Attendance

Requirements for class attendance & make-up exams, assignments, and other work are consistent with UF policies found at: <https://catalog.ufl.edu/ugrad/current/regulations/info/attendance.aspx>.

## In-Class Recording

**Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled.** The only allowable purposes are: (1) for personal educational use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor. **A ‘class lecture’ is:** an educational presentation intended to inform or teach enrolled students about a particular subject, including any instructor-led discussions that form part of the presentation, and delivered by any instructor hired or appointed by the University, or by a guest instructor, as part of a UF course. **A class lecture does not include:** lab sessions, student presentations, clinical presentations such as patient history, academic exercises involving solely student participation, assessments (quizzes, tests, exams), field trips, private conversations between students in the class or between a student and the faculty or lecturer during a class session. **Publication without permission of the instructor is prohibited. To ‘publish’ means:** to share, transmit, circulate, distribute, or provide access to a recording, regardless of format or medium, to another person (or persons), including

but not limited to another student within the same class section. Additionally, a recording, or transcript of a recording, is considered published if it is posted on or uploaded to, in whole or in part, any media platform, including but not limited to social media, book, magazine, newspaper, leaflet, or third-party note/tutoring services. ***A student who publishes a recording without written consent may be subject to a civil cause of action instituted by a person injured by the publication and/or discipline under UF Regulation 4.040 Student Honor Code and Student Conduct Code.***

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# UF Student Resources

## Health, Safety, & Wellness

**Wellness and Mental Health:** Students experiencing crises or personal problems that interfere with their general well-being are encouraged to utilize the university's counseling resources. The Counseling & Wellness Center provides confidential counseling services at no cost for currently enrolled students. Resources are available on campus for students having personal problems or lacking clear career or academic goals, which interfere with their academic performance. Visit <https://one.ufl.edu/whole-gator/discover> for resources that are designed to help you thrive physically, mentally, and emotionally at UF.

- **University Counseling & Wellness Center:** 3190 Radio Road, 352-392-1575. They can provide Counseling Services, Groups and Workshops, Outreach and Consultation, a Self-Help Library, and Wellness Coaching. <http://www.counseling.ufl.edu/>.
- **U Matter, We Care:** If you or someone you know is in distress, please contact [umatter@ufl.edu](mailto:umatter@ufl.edu), 352-392-1575, or visit U Matter, We Care website ([www.umatter.ufl.edu/](http://www.umatter.ufl.edu/)) to refer or report a concern and a team member will reach out to the student in distress.

**Student Health Care Center:** Call 352-392-1161 for 24/7 information to help you find the care you need, or visit the Student Health Care Center website.

**University Police Department:** Visit UF Police Department website or call 352-392-1111 (or 9-1-1 for emergencies).

**UF Health Shands Emergency Room / Trauma Center:** For immediate medical care call 352- 733-0111 or go to the emergency room at 1515 SW Archer Road, Gainesville, FL 32608; Visit the UF Health Emergency Room and Trauma Center website.

**Field and Fork Pantry:** The Hitchcock Pantry can provide food and toiletries for students experiencing food insecurity. <https://pantry.fieldandfork.ufl.edu/>.

## Academic Services

**E-learning technical support:** Contact the UF Computing Help Desk: (352) 392-4357 or [helpdesk@ufl.edu](mailto:helpdesk@ufl.edu).

**The Writing Studio** Help brainstorming, formatting, and writing papers. Daytime (9:30am-3:30pm): 2215 Turlington Hall, 352-846-1138. Evening (5:00pm-7:00pm): 1545 W University Avenue (Library West, Rm. 339).

**Career Connections Center** Reitz Union Ste 1300, (352) 392-1601. Career assistance & counseling services.

**Library Support** Various ways to receive assistance with respect to using the libraries or finding resources. Call 866-281-6309 or email [ask@ufl.libanswers.com](mailto:ask@ufl.libanswers.com) for more information.

**Academic Resources** 1317 Turlington Hall, Call (352) 392-2010 or to make a private appointment: 352-392-6420. Email contact: [teaching-center@ufl.edu](mailto:teaching-center@ufl.edu). General study skills and tutoring.

**Student Success Initiative** <http://studentsuccess.ufl.edu>.

**Student Complaints** Office of the Ombuds <https://ombuds.ufl.edu/>; Visit the Complaint Portal webpage <https://ombuds.ufl.edu/complaint-portal/> for more information.

**Enrollment Management Complaints (Registrar, Financial Aid, Admissions)** View the Student Complaint Procedure webpage for more information.

**UF Student Success Initiative** Visit <https://studentsuccess.ufl.edu/> for resources that support your success as a UF student.



# Assigned Reading & Viewing

## ! Important

Topics & readings are subject to change based on student interests and current events; changes will be announced via Canvas. *Please read or watch all assigned material before class.*

## Week 1

### Thursday: Course Introduction

- **Read:** no readings

## Week 2

### Tuesday: Historical Narratives

**Goal:** To introduce and discuss how historical narratives by the first Europeans to visit the tropics have shaped contemporary perceptions of tropical rain forests and the colonial roots of tropical biology

- **Read:** Excerpts from Historical Narratives (6 pages: [link](#))

### Thursday: Historical Narratives (cont.)

- **Read:** No readings

## Week 3

### Tuesday: Rain Forest Imagery in Art & Literature

**Goal:** To see how the different depictions of the tropics and tropical biodiversity from early European explorers are reflected in art and literature.

- **Read:** Excerpts from Literature/Poetry (5 pages: [link](#)) and Images of artwork (4 pages: [link](#))

### Thursday: The Rain Forest in Pop Culture

**Goal:** To compare the use and presentation of rain forest images by the private sector and in different forms of popular culture, including the film and music industries, and to evaluate how these depictions influence perceptions of tropical countries & people.

- **Read:** Jolly, Priscilla. 2021. 'Godzilla vs. Kong': Monster movies evoke adventure but also 'dangers' of tropics. *The Conversation*. ([link](#)) (4 pages)

## Week 4

### Tuesday: What is a Rain Forest?

**Goal:** To learn the different ways biologists define the "tropics" and how the structure and dynamics of tropical rain forests differ from those of forests in other parts of the world.

- **Watch:** Why Does Earth Have Deserts? ([link to video](#)); 2 min long) and
- **Watch:** BBC Planet Earth: Jungles. 2006. [Watch online](#) (access with your Gatorlink, 51 min).

## Week 5

### Thursday: Where is a Rain Forest?

**Goal:** To understand the geological history of tropical rain forests, how climate, fire, and geological history drive the tipping point between forests and savannas, and how this biogeographic, geological, and climatic history shaped the evolution of tropical plants and animals.

- **Read:** No readings.

## Week 5

### Tuesday: Patterns of Biodiversity

**Goal:** To observe and catalog the diversity of plant and animal life forms that can be found in rain forests, to quantify the local patterns of species richness and abundance in a tropical forest, and compare these patterns with those in the temperate zone.

- **Watch:** CBS News. Inside the Smithsonian's tropical forest lab. ([link](#), watch 5:35)
- **Read:** Butler, Rhett. 2015. How many tree species are found in the world's rainforests? *Mongabay* ([read online](#), 5 min.)

### Thursday: Patterns of Biodiversity, continued

**Goal:** To understand global patterns of species richness and how these vary from the tropics to the temperate zone.

- **Read:** Matthew Earl Boone and Mathieu Basille. Using iNaturalist to contribute your nature observations to science. UF EDIS Document WEC413: <https://journals.flvc.org/edis/article/view/107698/110114> (4 pages) and then familiarize yourself with the iNaturalist website: <https://www.inaturalist.org/>

## Week 6

### Tuesday: The Origins of Tropical Biodiversity

**Goal:** To review hypothesized mechanisms for the origins of tropical diversity and the role of interspecific interactions in the (co)evolution and diversification of tropical biodiversity

### Thursday: The maintenance of tropical biodiversity

**Goal:** To review the biotic and abiotic mechanisms in tropical rain forests that permit the coexistence of so many species

## Week 7

### Tuesday: Humans in Rain Forests

**Goal:** To understand the history of human occupation of rain forests including the contemporary demographic transition from rural to urban occupation; to review the different ways in which humans have historically modified rain forests and how this has shaped current rain forest biodiversity.

- **Watch:** Kristine Allington, Michael Amundson, Linithd Aparicio, & Caitlin Saks (Producers), & Townsley, G. (Director). (2023). Ancient Builders of the Amazon. [Video/DVD] Public Broadcasting Service. Retrieved from <https://video.alexanderstreet.com/watch/ancient-builders-of-the-amazon>
- **Read:** Hunt, Chris and Premathilake, Rathnasiri. 2018. "Prehistoric people started to spread domesticated bananas across the world 6,000 years ago." *The Conversation*. ([read online](#), 5 min).

### Thursday: Humans in Rain Forests, continued

- **Read:** No readings

## Week 8

### Tuesday: Paradox of Luxuriance

**Goal:** *To understand how such a productive biome can be built on such low-quality soils, and explore the implications of this “Paradox of Luxuriance”.*

- **Watch:** Anthony Bourdain’s Parts Unknown: Congo (S1E8) [[link to video](#), 50 min long].
- **Read:** Brooks, David. 2010. The Messiah Complex. *The New York Times*. ([link to editorial](#), 5 min. read))

### Thursday: Jungle Film Festival

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# Student Learning Objectives

B = Biological Sciences, Q2 = Quest 2

## CONTENT

*Students demonstrate competence in the terminology, concepts, theories, and methodologies used within the discipline(s)*

- (1) Describe the evolutionary and ecological factors underlying the distribution of biodiversity in tropical rain forests. *Assessments:* In-class activities. (B).
- (2) Distinguish (a) the different ways humans use and alter rain forests, and (b) the social, economic, and biological consequences of these activities. *Assessments:* In-class activities, Reflective Essay. (B, Q2)
- (3) Explain how genetics, remote sensing, computational tools, and other scientific developments have advanced research on the ecology and evolution of rain forest biota. *Assessments:* In-class activities, Analytic Essay. (B).
- (4) Examine the historical factors that have shaped cultural perceptions of rain forests. *Assessments:* In-class activities, Movie Reviews. (Q2).
- (5) Analyze how narratives about rain forests are reflected in different types of contemporary cultural expression. *Assessments:* In-class activities, Movie Reviews. (Q2).
- (6) Distinguish between the primary drivers of forest loss and how they vary geographically. *Assessments:* In-class activities, Analytic Essay. (B, Q2)
- (7) Critique different proposed mechanisms for rain forest conservation. *Assessments:* In-class activities, Analytic Essay. (B, Q2).

## CRITICAL THINKING

*Students carefully and logically analyze information from multiple perspectives and develop reasoned solutions to problems within the discipline(s).*

- (1) Test hypotheses regarding trends in deforestation and how forest regeneration varies geographically with quantitative data on forest cover and use news stories, primary literature, and other sources to propose mechanisms responsible for the patterns observed. *Assessments:* In-class activities, Analytic Essay (B, Q2).
- (2) Analyze qualitative data on rural-urban migration and human demographic shifts in tropical countries and assess the potential implications of the results for conservation and broader societal issues. *Assessments:* In-class activities, Analytic Essay (B, Q2).
- (3) Explain the relationship between deforestation and the global climate cycle. Compare alternative policy pathways for reducing CO<sub>2</sub> emissions using the En-ROADS global climate simulator. *Assessments:* In-class activities, Reflective Essay (B, Q2).



## COMMUNICATION

*Students communicate knowledge, ideas and reasoning clearly and effectively in written and oral forms appropriate to the discipline(s)*

- (1) Generate graphical analyses of quantitative data used to study rain forest biology and conservation (Q2, B). *Assessments:* In-class activities, Analytic Essay.
- (2) Compose summaries of research on multidisciplinary questions relevant to tropical forest biology and conservation based on logical arguments (Q2, B). *Assessments:* In-class activities, Analytic Essay.
- (3) Develop audience-specific content and materials with which to communicate an important issue related to tropical forest biology and conservation. *Assessments:* In-class activities, Reflective Essay, Movie Reviews, Analytic Essay, Final Project (Q2).

## CONNECTION

*Students connect course content with meaningful critical reflection on their intellectual, personal, and professional development at UF and beyond.*

- (1) Examine the cultural, economic, and historical experiences of people in rain forest countries, how this compares with our preconceived notions of the same, and the consequences of this disparity for our scientific understanding of rain forests and their conservation. *Assessments:* In-class activities, Movie Reviews, Final Project (Q2).
- (2) Assess the feedbacks between (a) the actions of individuals, governments, and the private sector, (b) global economic, social, and political conditions, and (c) the status rain forests and the global climate cycle. *Assessments:* In-class activities, Reflective Essay, Analytic Essay (Q2).
- (3) Appraise the ubiquity of tropical forest products in daily life, the role and impact of the global commodity chains that make this possible, and the consequences of consumer behavior for forest conservation and socioeconomic sustainability. *Assessments:* In-class activities, Final Project (Q2).
- (4) Analyze the local, regional, and global ecosystem services provided by tropical rain forests, how these vary geographically, and some of the cultural and ecological factors responsible for these differences. *Assessments:* In-class activities, Reflective Essay, Analytic Essay, Final Project (Q2)